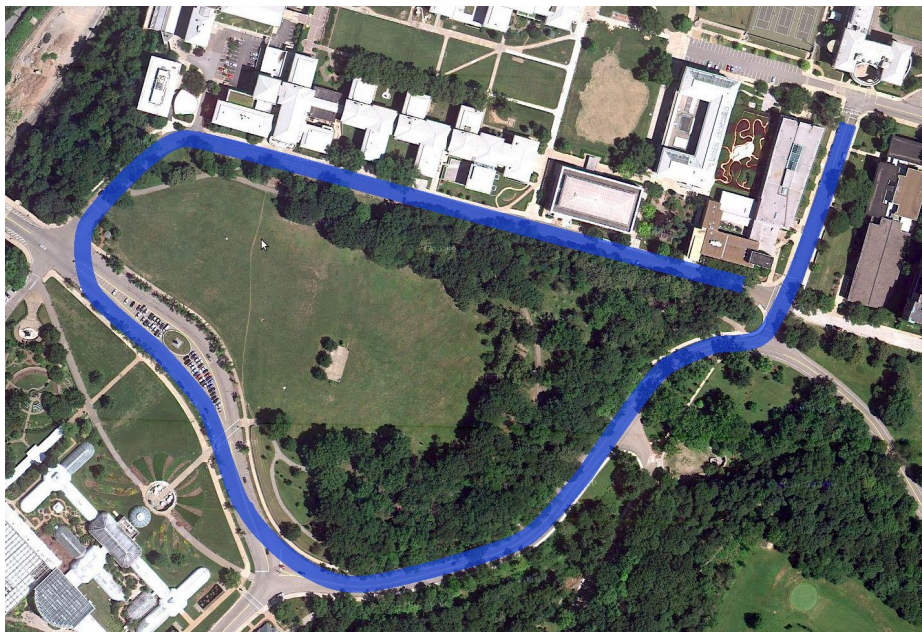




Reconstructing Environment and Tracking from Video

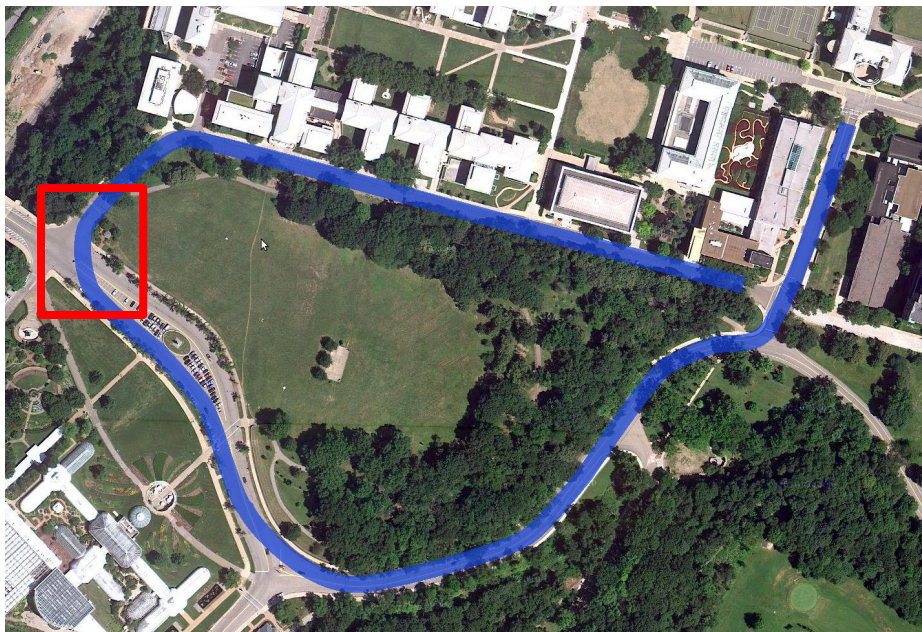
Riley Larget

Problem - The Race



- Many years of video
- What can we extract?
 - What's the topology?
 - Different teams drive slightly different paths
- Good sample problem to test techniques

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Video

CIA Mens B Spirit Womens B

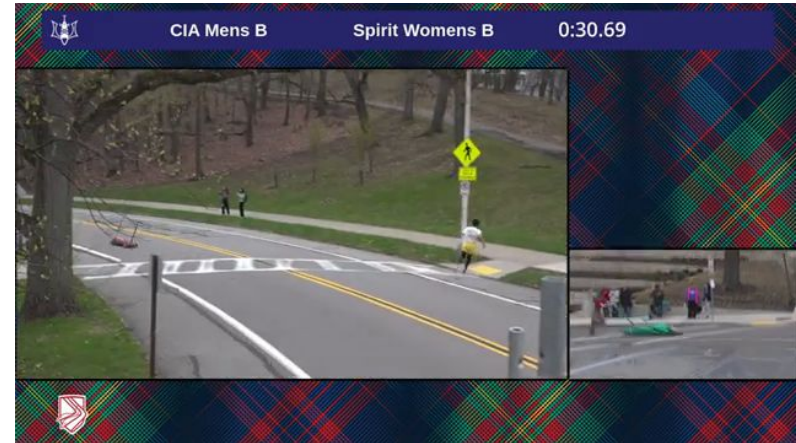
Roadrunner
Built: 2022

CIA Men's B

Driver	Amber Chang
Hill 1	Gavin Lin
Hill 2	Aiken Xie
Hill 3	Anson Li
Hill 4	Jordan Fon Sing
Hill 5	Oliver Graham



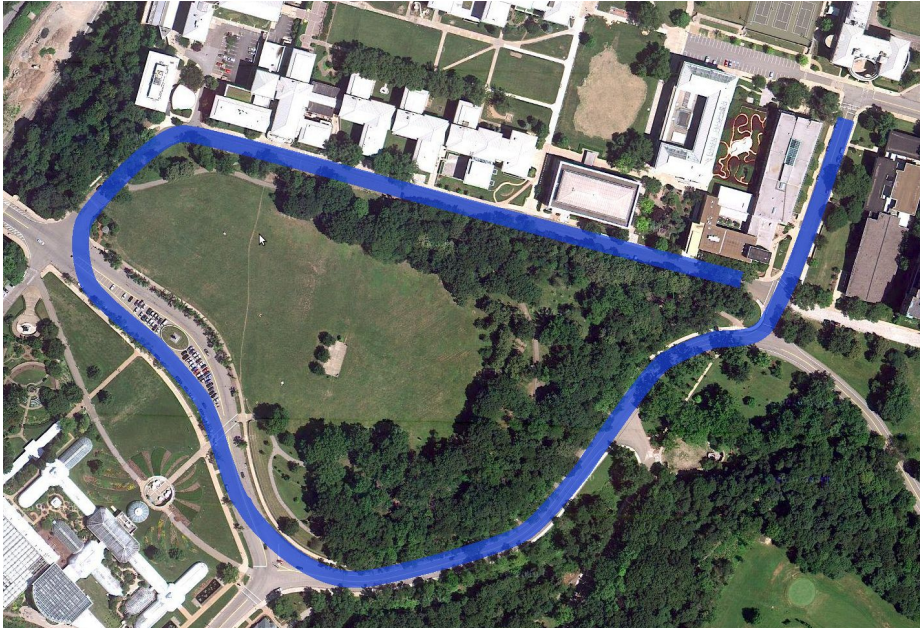
Video



Video

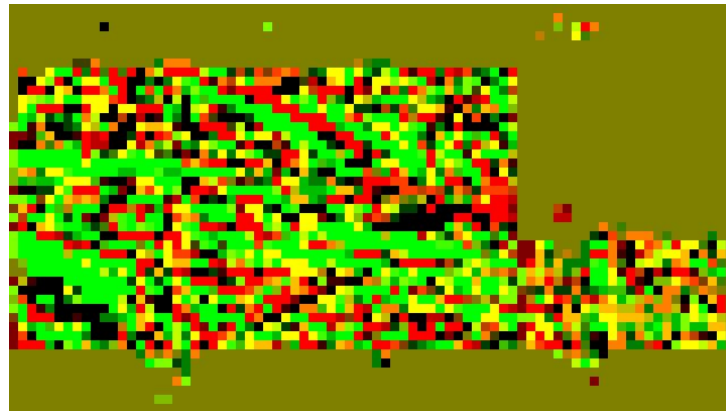


The Theory

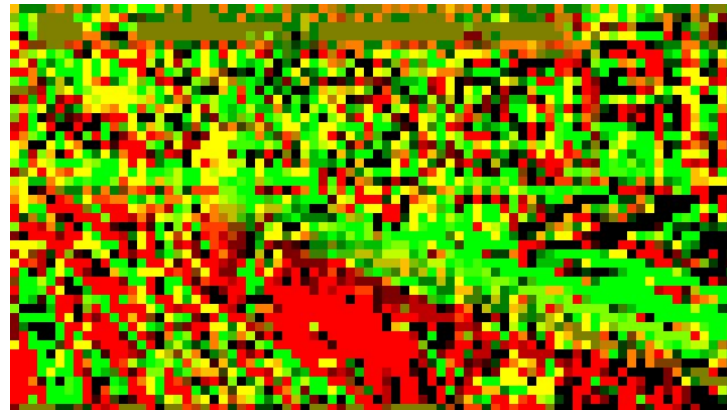


- Lots of similar images
- We can do 3-D stereo calculations between frames
- We can extract motion from optical flow between sequential frames

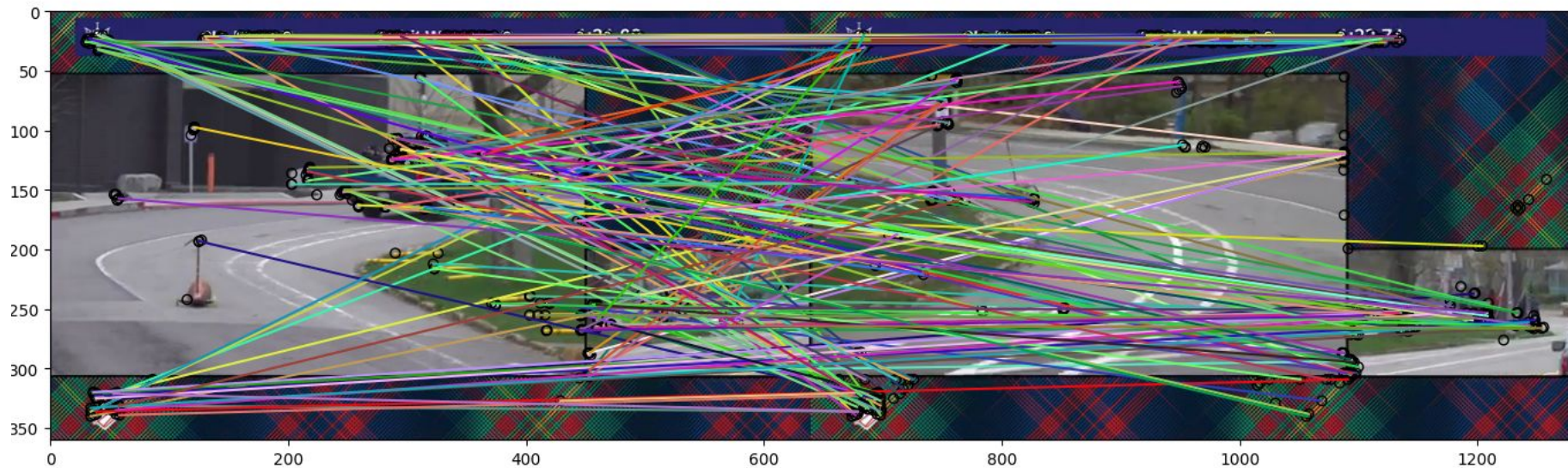
Optical Flow - in practice



Optical Flow - in practice



Naive feature extraction (ORB)



Video Extraction Problems

- Parts of the videos are not the race
- Overlays
- Multi-cam views
- Cameras zoom, pan, and cut

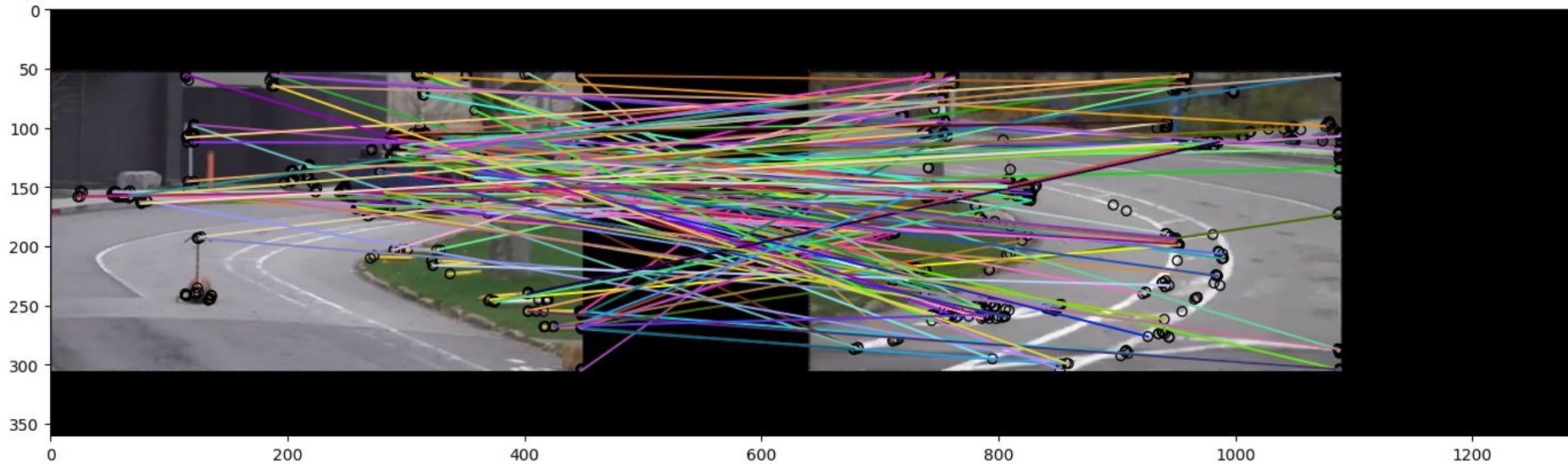


Video Extraction Solutions

- The race has a timer
- Overlays are in the same places
- Multi-cam uses the same places
 - Multi-cam has dividing lines
- Uncalibrated camera techniques
 - Background hard to isolate



Masked feature extraction (ORB)





Future work

- Continue extraction from matched pairs to 3-D points
- Extract vehicle and people from background environment
 - Possibly average background across races for the same section of the course
- Apply proven techniques to all videos